Solution of Homework 2

Note. This homework is due in class on February 20, 2026. Please show detailed steps to explain how you get your solution. Simply providing a final answer may not warrant full credit.

Question 1 [43503/53903] Compute singular value decomposition (SVD) for the following matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix}.$$

Solution. Computing $\mathbf{A}^*\mathbf{A}$ (notice, for real matrix $\mathbf{A}^* = \mathbf{A}^{\mathrm{T}}$) and its eigenvalues, we have

$$\mathbf{A}^*\mathbf{A} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

The characteristic polynomial of $\mathbf{A}^*\mathbf{A}$ is $\det(\mathbf{A}^*\mathbf{A} - \lambda\mathbf{I}) = -(\lambda - 3)^2(\lambda - 1)$. Thus, we get the eigenvalues of $\mathbf{A}^*\mathbf{A}$, as follows $\lambda_1 = \lambda_2 = 3$ and $\lambda_3 = 1$. Therefore, the singular values are

$$\sigma_1 = \sqrt{3}, \quad \sigma_2 = \sqrt{3}, \quad \text{and} \quad \sigma_3 = 1.$$

Next, let us compute eigenvectors of $\mathbf{A}^*\mathbf{A}$. For each λ_i $(i = 1, 2, 3)$, we solve $\mathbf{A}^*\mathbf{A}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$. We have

$$\boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \boldsymbol{v}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \boldsymbol{v}_3 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}.$$

For each singular value σ_i $(i = 1, 2, 3)$, the associated left singular vector satisfies $\boldsymbol{u}_i = \frac{1}{\sigma_i}\mathbf{A}\boldsymbol{v}_i$. Thus, we obtain

$$\boldsymbol{u}_1 = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \boldsymbol{u}_2 = \frac{1}{\sqrt{6}}\begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}, \quad \text{and} \quad \boldsymbol{u}_3 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

Therefore, we get $\mathbf{\Sigma} = \operatorname{diag}(\sqrt{3}, \sqrt{3}, 1)$ and

$$\mathbf{V} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \quad \text{and} \quad \mathbf{U} = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

Question 2 [43503/53903] Compute the SVD of the matrix $\mathbf{A} = \begin{pmatrix} -2 & 11 \\ -10 & 5 \end{pmatrix}$ in the form of $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$. Notice, the SVD is not unique, you only need to find one and validate your solution.

(a) What are the 1-norm, 2-norm, ∞-norm, and Frobenius norm of $\mathbf{A}$?

(b) Verify that $\det(\mathbf{A}) = \lambda_1\lambda_2$ and $|\det(\mathbf{A})| = \sigma_1\sigma_2$, where the λ_1, λ_2 are eigenvalues and σ_1, σ_2 are singular values of $\mathbf{A}$.

Solution. Using the similar procedure as in Question 1, we get the singular value decomposition of $\mathbf{A}$ is

$$\mathbf{A} = \underbrace{\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}}_{=\mathbf{U}} \underbrace{\begin{pmatrix} 10\sqrt{2} & 0 \\ 0 & 5\sqrt{2} \end{pmatrix}}_{=\mathbf{\Sigma}} \underbrace{\begin{pmatrix} -\frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{pmatrix}}_{=\mathbf{V}^*}.$$

(a) For the matrix norms, we have

- $\|\mathbf{A}\|_1 = \max\{|-2| + |-10|, |11| + |5|\} = 16$
- $\|\mathbf{A}\|_\infty = \max\{|-2| + |11|, |-10| + |5|\} = 15$
- $\|\mathbf{A}\|_2 = \sigma_1 = 10\sqrt{2}$
- $\|\mathbf{A}\|_{\mathrm{F}} = \sqrt{\sigma_1^2 + \sigma_2^2} = 5\sqrt{10}$

(b) The determinant of $\mathbf{A}$ is $\det(\mathbf{A}) = (-2)(5) - (11)(-10) = 100$. To compute the eigenvalue of $\mathbf{A}$, we have the characteristic equation

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} -2-\lambda & 11 \\ -10 & 5-\lambda \end{vmatrix} = \lambda^2 - 3\lambda + 110 = 0.$$

Therefore, the two eigenvalues of matrix $\mathbf{A}$ are

$$\lambda_1 = \frac{3 + \sqrt{391}i}{2} \quad \text{and} \quad \lambda_2 = \frac{3 - \sqrt{391}i}{2}.$$

Thus, $\lambda_1\lambda_2 = 100$, namely the $\det(\mathbf{A}) = \lambda_1\lambda_2$ holds. Recall $\sigma_1 = 10\sqrt{2}$ and $\sigma_2 = 5\sqrt{2}$, we have $\sigma_1\sigma_2 = 100$. Therefore, the $|\det(\mathbf{A})| = \sigma_1\sigma_2$ holds.

Question 3 [43503/53903] Show the following statements hold.

(a) Given $\mathbf{A} \in \mathbb{C}^{m\times n}$ with $m \geq n$, show that $\mathbf{A}^*\mathbf{A}$ is nonsingular if and only if $\mathbf{A}$ has full rank.

(b) Let $\mathbf{P} \in \mathbb{C}^{m\times m}$ be a nonzero projector. Show that $\|\mathbf{P}\|_2 \geq 1$. In addition, show that if $\mathbf{P}$ is an orthogonal projector, then $\|\mathbf{P}\|_2 = 1$.

Solution. (a) "$\Rightarrow$" Given that $\mathbf{A}^*\mathbf{A} \in \mathbb{C}^{n\times n}$ is nonsingular, let us show $\mathbf{A} \in \mathbb{C}^{m\times n}$ $(m \geq n)$ is full rank by applying contradiction argument. Assume $\operatorname{rank}(\mathbf{A}) < n$, then the linear system $\mathbf{A}\boldsymbol{x} = \mathbf{0}$, namely

$$\underbrace{\begin{bmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \cdots & \boldsymbol{a}_n \end{bmatrix}}_{=\mathbf{A}} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \cdots + x_n\boldsymbol{a}_n = \mathbf{0},$$

has nonzero solution $\boldsymbol{\xi} \in \mathbb{C}^n$ (the columns of $\mathbf{A}$ are linearly dependent). Left multiplying $\mathbf{A}^*$, we obtain $\mathbf{A}^*\mathbf{A}\boldsymbol{\xi} = \mathbf{0}$. Notice, $\boldsymbol{\xi} \neq \mathbf{0}$ implies that $\mathbf{A}^*\mathbf{A}$ has a nontrivial nullspace and is therefore singular. Contradict to $\mathbf{A}^*\mathbf{A}$ is nonsingular. Hence, $\mathbf{A}$ must have full rank.

"$\Leftarrow$" Given that $\mathbf{A}$ has full rank, let us show $\mathbf{A}^*\mathbf{A}$ is nonsingular. This is equivalent to show the homogeneous linear system $\mathbf{A}^*\mathbf{A}\boldsymbol{x} = \mathbf{0}$ has only zero solution. Assume $\boldsymbol{\xi} \in \mathbb{C}^n$ is a solution, i.e. $\mathbf{A}^*\mathbf{A}\boldsymbol{\xi} = \mathbf{0}$. Then, we have $\|\mathbf{A}\boldsymbol{\xi}\|_2^2 = \boldsymbol{\xi}^*\mathbf{A}^*\mathbf{A}\boldsymbol{\xi} = 0$, which implies $\mathbf{A}\boldsymbol{\xi} = \mathbf{0}$. Notice, $\mathbf{A}$ has full rank (the columns of $\mathbf{A}$ is linearly independent). Therefore

$$x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \cdots + x_n\boldsymbol{a}_n = \begin{bmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \cdots & \boldsymbol{a}_n \end{bmatrix} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{=\,\xi} = \mathbf{0},$$

gives $\boldsymbol{\xi} = \mathbf{0}$. Thus, $\mathbf{A}^*\mathbf{A}$ is thus nonsingular.
(b) The matrix $\mathbf{P}$ is a nonzero projector, then range($\mathbf{P}$) is nontrivial. Pick a vector $\boldsymbol{y} \neq \mathbf{0}$ in range($\mathbf{P}$), namely $\mathbf{P}\boldsymbol{y} = \boldsymbol{y}$. We have

$$\|\mathbf{P}\|_2 = \sup_{\boldsymbol{x}\neq\mathbf{0}} \frac{\|\mathbf{P}\boldsymbol{x}\|_2}{\|\boldsymbol{x}\|_2} \geq \frac{\|\mathbf{P}\boldsymbol{y}\|_2}{\|\boldsymbol{y}\|_2} = \frac{\|\boldsymbol{y}\|_2}{\|\boldsymbol{y}\|_2} = 1.$$

Given that $\mathbf{P}$ is an orthogonal projector, i.e., $\mathbf{P}^2 = \mathbf{P}$ and $\mathbf{P}^* = \mathbf{P}$. Then, the eigenvalues of $\mathbf{P}$ are either 0 or 1. Notice $\mathbf{P}$ is hermitian. The 2-norm of a hermitian matrix equals the maximum absolute value of its eigenvalues, we get $\|\mathbf{P}\|_2 = 1$.

Question 4 [43503/53903] Compute the reduced and full QR factorization of the following matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Solution. The reduced QR factorization is as follows

$$\mathbf{A} = \widehat{\mathbf{Q}}\widehat{\mathbf{R}}, \quad \text{where} \quad \widehat{\mathbf{Q}} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \end{pmatrix} \quad \text{and} \quad \widehat{\mathbf{R}} = \begin{pmatrix} \sqrt{2} & \sqrt{2} \\ 0 & \sqrt{3} \end{pmatrix}.$$

The full QR factorization is as follows

$$\mathbf{A} = \mathbf{Q}\mathbf{R}, \quad \text{where} \quad \mathbf{Q} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{pmatrix} \quad \text{and} \quad \mathbf{R} = \begin{pmatrix} \sqrt{2} & \sqrt{2} \\ 0 & \sqrt{3} \\ 0 & 0 \end{pmatrix}.$$

Question 5 [53903]

(a) Let $\mathbf{A}$ be an $m \times n$ matrix ($m \geq n$), and let $\mathbf{A} = \widehat{\mathbf{Q}}\widehat{\mathbf{R}}$ be a reduced QR factorization. Show that $\mathbf{A}$ has rank n if and only if all the diagonal entries of $\widehat{\mathbf{R}}$ are nonzero.

(b) Implement your own code to compute the QR factorization of the following matrix and compare your result with MATLAB build-in function.

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 7 \\ 4 & 2 & 3 \\ 4 & 2 & 2 \end{pmatrix}.$$

Solution. (a) Recall the reduced QR factorization of matrix $\mathbf{A} = \widehat{\mathbf{Q}}\widehat{\mathbf{R}}$, where $\widehat{\mathbf{Q}}$ is $m \times n$ with orthonormal columns (linear independent columns) and $\widehat{\mathbf{R}}$ is $n \times n$ upper triangular.
"$\Leftarrow$" The diagonal entries of $\widehat{\mathbf{R}}$ are nonzero, implies $\widehat{\mathbf{R}}$ is invertible. Assume, for contradiction, $\mathbf{A} = [\boldsymbol{a}_1 \ \boldsymbol{a}_2 \ \cdots \ \boldsymbol{a}_n]$ is not full rank. Then, there exist not all zero $x_1, x_2, \cdots, x_n$, such that

$$\underbrace{\begin{bmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \cdots & \boldsymbol{a}_n \end{bmatrix}}_{=\mathbf{A}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_{=\boldsymbol{x}} = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \cdots + x_n\boldsymbol{a}_n = 0.$$

The reduced QR factorization gives $\mathbf{A} = \widehat{\mathbf{Q}}\widehat{\mathbf{R}}$. Let $\boldsymbol{y} = \widehat{\mathbf{R}}\boldsymbol{x}$. Then $\boldsymbol{y} \neq \mathbf{0}$, otherwise $\widehat{\mathbf{R}}\boldsymbol{x} = \mathbf{0}$ with nonsingular $\widehat{\mathbf{R}}$ implies $\boldsymbol{x} = \mathbf{0}$. Contradict to $x_1, x_2, \cdots, x_n$ are not all zero. Therefore, $\mathbf{A}\boldsymbol{x} = \widehat{\mathbf{Q}}\widehat{\mathbf{R}}\boldsymbol{x} = \widehat{\mathbf{Q}}\boldsymbol{y}$

$$\underbrace{\begin{bmatrix} \boldsymbol{q}_1 & \boldsymbol{q}_2 & \cdots & \boldsymbol{q}_n \end{bmatrix}}_{=\widehat{\mathbf{Q}}} \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}}_{=\boldsymbol{y} \neq \mathbf{0}} = y_1\boldsymbol{q}_1 + y_2\boldsymbol{q}_2 + \cdots + y_n\boldsymbol{q}_n = 0,$$

implies $\boldsymbol{q}_1, \boldsymbol{q}_2, \cdots, \boldsymbol{q}_n$ are linearly dependent. Contradict. Thus $\mathbf{A}$ is full rank.
"$\Rightarrow$" The matrix $\mathbf{A}$ is full rank, namely $\operatorname{rank}(\mathbf{A}) = n$. Notice, the matrix $\widehat{\mathbf{Q}}$ is $m \times n$ with orthonormal columns, namely $\operatorname{rank}(\mathbf{Q}) = n$. From the reduced QR factorization $\mathbf{A} = \widehat{\mathbf{Q}}\widehat{\mathbf{R}}$, we have

$$n = \operatorname{rank}(\mathbf{A}) \leq \min(\operatorname{rank}(\widehat{\mathbf{Q}}), \operatorname{rank}(\widehat{\mathbf{R}})) = \min(n, \operatorname{rank}(\widehat{\mathbf{R}})) \leq \operatorname{rank}(\widehat{\mathbf{R}}).$$

Notice, the matrix $\widehat{\mathbf{R}}$ is an n-by-n matrix, we have $\operatorname{rank}(\widehat{\mathbf{R}}) \leq n$. From $\operatorname{rank}(\widehat{\mathbf{R}}) \geq n$ above, we get $\operatorname{rank}(\widehat{\mathbf{R}}) = n$, which implies $\widehat{\mathbf{R}}$ is nonsingular. We have

$$0 \neq \det(\widehat{\mathbf{R}}) = \begin{vmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ & r_{22} & \cdots & r_{2n} \\ & & \ddots & \vdots \\ & & & r_{nn} \end{vmatrix} = r_{11}r_{22}\cdots r_{nn}.$$

Therefore, all the diagonal entries of $\widehat{\mathbf{R}}$ are nonzero.
(b) Here is the Matlab code for QR factorization.

Use classical Gram–Schmidt QR factorization

```
%% Classical Gram-Schmidt QR factorization
% Input:
%       A : m-by-n real matrix
%
% Output:
%       Q : m-by-n matrix with orthonormal columns
%       R : n-by-n upper triangular matrix

function [Q,R] = clgs(A)
  [m,n] = size(A);

  Q = zeros(m,n);
  R = zeros(n,n);

  for j = 1:n
    v = A(:,j);

    for i = 1:j-1
      R(i,j) = Q(:,i)'*A(:,j);
      v = v - R(i,j)*Q(:,i);
    end

    R(j,j) = norm(v);
    Q(:,j) = v/R(j,j);
  end
end
```

Use modified Gram–Schmidt QR factorization

```
%% Modified Gram-Schmidt QR factorization
% Input:
%       A : m-by-n real matrix
%
% Output:
%       Q : m-by-n matrix with orthonormal columns
%       R : n-by-n upper triangular matrix

function [Q,R] = mgs(A)
[m,n] = size(A);

Q = zeros(m,n);
R = zeros(n,n);
```

```
V = A;  % for i = 1 to n: v_i = a_i

  for i = 1:n
    R(i,i) = norm(V(:,i));
    Q(:,i) = V(:,i)/R(i,i);

    for j = i+1:n
      R(i,j) = Q(:,i)' * V(:,j);
      V(:,j) = V(:,j) - R(i,j)*Q(:,i);
    end
  end
end
```